Terminology Glossary

Extracted only from Chapter 7 terminology and assumptions.

Section	Term	Extracted text	Page
7.1 INTRODUCTION	Resource definition	A resource is any software structure that can be used by a process to advance its execution.	1
7.1 INTRODUCTION	Private/shared/exclusive resource	A resource dedicated to a particular process is said to be private, whereas a resource that can be used by more tasks is called a shared resource. A shared resource protected against concurrent accesses is called an exclusive resource.	1
7.1 INTRODUCTION	Critical section	A piece of code executed under mutual exclusion constraints is called a critical section.	1
7.1 INTRODUCTION	Blocked task	A task waiting for an exclusive resource is said to be blocked on that resource, otherwise it proceeds by entering the critical section and holds the resource.	1
7.1 INTRODUCTION	Semaphore	Operating systems typically provide a general synchronization tool, called a semaphore [Dij68, BH73, PS85], that can be used by tasks to build critical sections.	1
7.2 THE PRIORITY INVERSION PHENOMENON	Priority inversion	A priority inversion is said to occur in the interval [t3, t6], since the highest-priority task τ1 waits for the execution of lower-priority tasks (τ2 and τ3).	4
7.3 TERMINOLOGY AND ASSUMPTIONS	Bi	Bi denotes the maximum blocking time task τi can experience.	5
7.3 TERMINOLOGY AND ASSUMPTIONS	zi,k	zi,k denotes a generic critical section of task τi guarded by semaphore Sk.	5
7.3 TERMINOLOGY AND ASSUMPTIONS	Zi,k	Zi,k denotes the longest critical section of task τi guarded by semaphore Sk.	5
7.3 TERMINOLOGY AND ASSUMPTIONS	delta i,k	δi,k denotes the duration of Zi,k.	6
7.3 TERMINOLOGY AND ASSUMPTIONS	Nested critical sections	The critical sections used by any task are properly nested; that is, given any pair zi,h and zi,k, then either zi,h ⊂ zi,k, zi,k ⊂ zi,h, or zi,h ∩ zi,k = ∅.	6
7.6.1 PROTOCOL DEFINITION	Direct blocking	Direct blocking. It occurs when a higher-priority task tries to acquire a resource already held by a lower-priority task.	12
7.6.1 PROTOCOL DEFINITION	Push-through blocking	Push-through blocking. It occurs when a medium-priority task is blocked by a low-priority task that has inherited a higher priority from a task it directly blocks.	12
7.7.1 PROTOCOL DEFINITION	Ceiling blocking	Note that the Priority Ceiling Protocol introduces a third form of blocking, called ceiling blocking, in addition to direct blocking and push-through blocking caused by the Priority Inheritance Protocol.	25
Chapter 7	Preemption level	Resource Access Protocols 235 PREEMPTION LEVEL Besides a priority pi, a task τi is also characterized by a preemption level πi.	31
Chapter 7	Preemption level	The preemption level is a static parameter, assigned to a task at its creation time and as- sociated with all instances of that task.	31

Section	Term	Extracted text	Page
Chapter 7	System ceiling	238 Chapter 7 SYSTEM CEILING The resource access protocol adopted in the SRP also requires a system ceiling, Πs, de■ned as the maximum of the current ceilings of all the resources; that is, Πs =m a x k {CRk }.	34
Chapter 7	System ceiling	Using the de■nitions introduced in the previous paragraph, this is achieved by the following preemption test: SRP Preemption Test: A task is not permitted to preempt until its priority is the highest among those of all the tasks ready to run, and its preemption level is higher than the system ceiling.	34